Ball electrodes for the chest

complete CardioMon set is available for ₹ 15,000 (plus GST) on *www.amazon.in*, as well as through offline distribution channels. In the box, the main CardioMon ECG monitor and analyser device comes with 10 electrodes and an ECG cable.

Technical and ecosystem obstacles

Research-based innovation is time- and effort-dependent, and involves some risk in terms of deriving assured success. Sharma believes that this is where the Indian ecosystem becomes restricted. As a result, receiving funding also becomes difficult. He was denied funding for this research project, since investors believed that a single-person driving this project was too far-fetched. Finally, he had to bootstrap his project.

"The main concern is the hesitation of investors in taking the risk of investing and waiting for the fruits of research. It is a chicken-and-egg situation. Someone needs to take an initiative and persevere. Without that we would never have innovations that we use so commonly today," Sharma adds.

Familiarising himself with biomedical subject matter and learning the working principles of these devices also factored in.

From a design point of view, miniaturising the product design based on functionality and accuracy of a full-size ECG machine was difficult. Sharma recalls the first prototype to

Clamp electrodes for the limbs

be bulky and heavy.

Cost control was also challenging.

Anusandhan Technologies team has been working with doctors in and around Pune, Maharashtra, with this device for constant improvements. Sharma says, "Dr Pramod Narkhede of N.M. Wadia Institute of Cardiology has been a major contributor to the product lifecycle. He has been practically testing CardioMon for the last eight months, and has been giving substantial inputs towards its user friendliness and usefulness."

Addition of accessories such as vest and printing feature was also driven by feedback from doctors who tried the device.

The road ahead

Anusandhan Technologies plans to scale production, based on the market performance of CardioMon. The team is in the process of expanding its distribution channels through partners who have the technical know-how and experience with such biomedical devices.

As a feature upgrade, the team is ready to add support, so that results can be sent to doctors via a smartphone for their viewing and storage of records, and monitor progress. Bluetooth compatibility with desktops and laptops is also being looked into. The team has also filed for a patent for the vest.

"It is good to see the rising number of innovators in India. However, the path to cover is long. Besides Make in India initiative, the time has come to have a movement like Innovate in India, to relieve us from our dependencies on other nations for our consumer and defence needs," concludes Sharma.

Artificial Intelligence To Help Reduce Crime

India has consistently suffered from a high number of criminal activities and law violations over the years. Official records suggest that 70 per cent of these incidents involve repeat offenders. For the longest time this kind of data remained offline, even when it was spread across different zones and states. Identifying this major challenge law protectors of our nation deal with every day, Gurugram-based Staqu Technologies pledged to use technology to empower law enforcement.

Anurag Saini, Pankaj Sharma, Chetan Rexwal and Atul Rai, co-founders of Staqu, were the first to conceive the idea of a data analytics-based artificial intelligence (AI)-driven platform that can bring necessary data to Indian law administratives. The solution is called AI-Based Human Efface Detection, or simply ABHED.

Staqu team intended to utilise AI in digitising all the Big Data of criminal records, while extracting meaningful attributes to provide police forces with real-time information.

Rai, who is also chief executive officer at Staqu, says, "When we started Staqu, Indian police forces were utilising solutions provided by international countries such as Israel, Japan and Scandinavian countries. It